

Practical - 6

Practical Title: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

A. Theory

1. Is the schema normalized properly?

Yes. The database is divided into separate tables such as **Students, Events, and Registrations**. This avoids duplicate data and ensures that each table stores information related to a specific entity. The Registration table connects students and events.

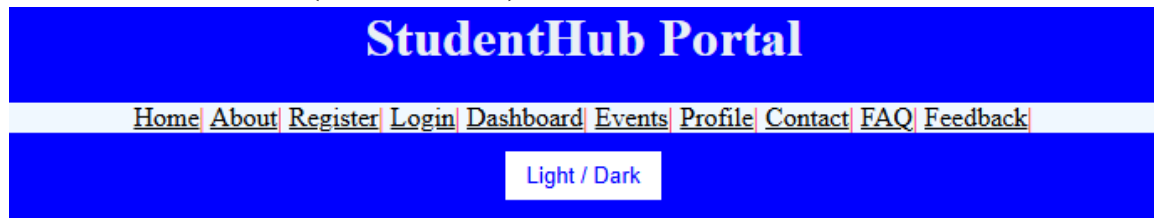
2. Are primary keys and foreign keys correctly defined?

Yes. Each table has a unique **primary key** to identify its records. The **foreign keys** in the Registration table refer to the Student and Event tables, maintaining the relationship and data integrity between the tables.

3. Is the database connection handled securely?

Yes. The PHP database connection uses **PDO** with exception-based error handling. Prepared statements can be used for database queries to prevent **SQL injection**. Database credentials should also be kept separate from other application code and should not be exposed to users.

Evidence of Work (Screenshots)



Upcoming Events

Web Development Workshop

Date: 2026-08-10
Venue: Seminar Hall
Category: Workshop

Hackathon

Date: 2026-08-13
Venue: Computer Lab
Category: Competition

Sports Day

Date: 2026-08-16
Venue: College Ground
Category: Sports

Sports Day

Date: 2026-08-16
Venue: College Ground
Category: Sports

Coding Contest

Date: 2026-08-19
Venue: Seminar Hall
Category: Competition

AI Seminar

Date: 2026-08-22
Venue: Computer Lab
Category: Seminar

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StudentHub Portal

[Home](#) | [About](#) | [Register](#) | [Login](#) | [Dashboard](#) | [Events](#) | [Profile](#) | [Contact](#) | [FAQ](#) | [Feedback](#)

Light / Dark

Upcoming Events

hackathon

All Categories ▼

Date: Oldest First ▼

Hackathon Date: 2026-08-13 Venue: Computer Lab Category: Competition
Hackathon Date: 2026-09-13 Venue: Seminar Hall Category: Competition
Hackathon Date: 2026-10-13 Venue: College Ground Category: Competition

Description:

- **Schema Normalization:**

The database is divided into separate tables for students, events, and registrations. This reduces data duplication and keeps the database organized.

- **Primary and Foreign Keys:**

Each table has a primary key that uniquely identifies every record. Foreign keys are used in the registrations table to connect students with events and maintain relationships between the tables.

- **Secure Database Connection:**

The PHP application uses PDO to establish a database connection. Error handling is used

to manage connection failures, and prepared statements can be used to protect database queries from SQL injection.

B. GitHub Repository Link:

<https://github.com/vatsal-1950/StudentHUB.git>

Challenges Faced

1. **Database design:** Deciding how to divide the StudentHub data into separate tables for students, events, and registrations was challenging.
2. **Relationships:** Understanding and correctly implementing relationships between tables using foreign keys required careful attention.
3. **Primary and foreign keys:** Defining the keys correctly and maintaining data integrity was a challenge.
4. **PHP-PDO connection:** Setting up the connection between PHP and MySQL and handling connection errors required debugging.
5. **Prepared statements:** Understanding how prepared statements are used for secure database operations was initially challenging.
6. **Testing:** Testing the database connection and checking whether the tables and relationships worked correctly required multiple checks.

Learning Outcome

After completing this practical, I learned how to:

1. **Design a normalized MySQL database** for a web application.
2. **Create tables using MySQL** and insert seed/test data.
3. **Define primary and foreign keys** to maintain relationships and data integrity.
4. **Create an ER model** to represent relationships between students, events, and registrations.
5. **Connect PHP with MySQL using PDO.**
6. **Handle database connection errors** using exception handling.
7. **Understand prepared statements** and their importance for secure database queries.
8. **Use phpMyAdmin and XAMPP** to create, manage, and test the database.

Faculty Remarks: (Keep this blank for faculty)